Mohammed Fahim

167, 6th Block, Mogappair West, Chennai, Tamil Nadu, India-600037·
mailmefahim14@gmail.com · + 91 (638) 248-3853.
LinkedIn | Portfolio | Github

SUMMARY

Software Engineer with 2 years of experience designing and delivering AI-powered, cloud-native, and serverless solutions on AWS. **AWS Certified Solutions Architect – Associate** and **AWS Certified Security – Specialty**, specializing in scalable automation, least-privilege cloud architecture, and real-time enterprise applications..

TECHNICAL SKILLS

- **Programming & Web Development:** Python, JavaScript, TypeScript, Node.js, React, Next.js, Vite, Tailwind CSS, Vercel, FastAPI, Flask, HTML5
- **AWS Cloud & Services:** AWS IAM, Lambda, S3, CloudFront, DynamoDB, EC2, Step Functions, SQS, EventBridge, CloudWatch, VPC Endpoints, Elastic Beanstalk, ALB, Auto Scaling Groups, AWS CloudFormation, AWS Config, AWS Secrets Manager
- **AI/ML Technologies:** AWS Bedrock, Amazon Titan Embeddings, OpenAI GPT, OpenAI Embeddings, Gemini AI, Pinecone, Scikit-learn, LlamaParse, Strands SDK
- **Database & Data Processing:** PostgreSQL, SQLAlchemy, RDS, MySQL, DynamoDB Streams, DynamoDB vector index, Redis Cloud, RediSearch, Pandas, ThreadPoolExecutor, Data Pipelines, ETL Processes
- **API Development & Integration:** RESTful APIs, OpenAPI 3.0, API Gateway, WebSocket, WebRTC, LAML/cXML, Gmail API, Slack API, SignalWire, Twilio, SMTP
- **DevOps & Infrastructure:** Git, GitHub, GitHub Actions, Docker, ECS Fargate, Playwright, Jenkins
- **Security & Compliance:** AWS Cognito User Pools, Amazon GuardDuty, OAuth 2.0, JWT, OTP, bcrypt, SSL/TLS

PROFESSIONAL EXPERIENCE

FairPAY Tech Works | AI & AWS Engineer — December 2024 – Present

- Developed an AI-powered Fleet Operations Intelligence Agent using **AWS Bedrock Agents**, **Lambda**, **DynamoDB**, and **API Gateway WebSocket,** implementing 9 API endpoints via **OpenAPI 3.0 Schema**, **35+** dashboard filters, **Redis Cloud** semantic search with **RediSearch**, and **DynamoDB Streams** for live indexing, delivering natural language fleet intelligence for vehicle, financial, insurance, and maintenance operations.
- Architected a business communication platform using **FastAPI**, **PostgreSQL**, **SignalWire**, and **S3**/**CloudFront**, implementing SMS/MMS, **WebRTC** voice calls with **LAML**/**cXML**, **JWT** + **OTP** auth with **bcrypt** and **RBAC**, and 50+ RESTful endpoints, enhanced with **AI chat auto-reply** using **OpenAI GPT**, **Embeddings**, and **Scikit-learn** for semantic routing and FAQ automation, achieving **60% faster** response times with **ThreadPoolExecutor**.
- Engineered an enterprise data intelligence and workflow automation platform with **AWS Lambda**, **PostgreSQL**, and **SQS**, integrating **LlamaParse** (**95% accuracy**) and **Gemini AI** for automated invoice/payment extraction and validation, **Slack**-based approval notifications, **Playwright**-powered insurance quote retrieval, and **AWS Bedrock agents** for natural language analytics, delivering real-time financial dashboards with end-to-end form automation, tracking, and documentation systems.
- Developed an **AI**-powered vehicle intelligence platform using **Next.js**, **Lambda**, **API Gateway**, **DynamoDB vector index**, **ECS Fargate**, **Step Functions**, **EventBridge**, and **SQS**, with **Amazon Bedrock (Titan embeddings, Grok)** and **Gemini** for semantic search, ranking, and pricing, secured through **Cognito**, **IAM**, **Secrets Manager**, and **VPC endpoints.**
- Orchestrated a real-time communication platform integrating **Gmail API** and **WebSocket** services, enabling email-to-SMS functionality, live **Google Voice** messaging, and conversation tracking with **DynamoDB**-backed persistence, delivering **99.9%** uptime for reliable enterprise communication.
- Designed an AI-powered business chatbot using **AWS Bedrock** and **Pinecone** vector storage, integrating the **Strands Agent** framework for factual retrieval, real-time chat, and admin correction workflows, while delivering scalable support via **React** and **WebSocket**-based interfaces.

CERTIFICATION

AWS Certified Solutions Architect – Associate — **Mar 2025 – Mar 2028**
AWS Certified Security - Specialty — **Oct 2026 – Oct 2029**

EDUCATION

Sathyabama University — **Chennai, India**
Bachelor of Engineering Computer Science; — **Sept 2021 – June 2025**